



Project Report - Fundamentals of AIML

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Problem Statement

I would like to introduce you to a rather difficult yet most faced problem in the world especially by business owners, "Retail Demand".

Predicting demand is a very intricate game as one store could be starving for customers on a daily basis while on the other hand, some stores get their entire inventory wiped in 8 hours. Sometimes this is not the store's fault, it can be the customer's behaviour as well. Customers usually judge a store based on its location, size, quality of life facilities (such as parking, snacking points), frequency and magnitude of discounts given etc.

Customer behaviour also heavily depends on the day of purchase. If it's a holiday with calm weather, we can safely assume that the store will get more customers but, if it's raining on a working day (Monday to Friday), we better keep our expectations low.

I decided it was time to sit down, figure Demand out, and finally predict their next move. This project report details exactly how I did it.

Approach Taken: Converting Human Behavior into Maths

For a long time, data scientists have been predicting sales with the help of past receipts of the customers but this wasn't too effective as we know that predicting demand can be very chaotic.

So I decided that it is time to change this, but to do so I would have to dive deeper than just past receipts, I would have to understand the behavior of the customers and what affects it. And so I did.

I did this by feeding the LightGBM regressor with past sales data but instead of just feeding it raw data, I made it make sense for the regressor by converting dates into a format that the algorithm

can understand. I also combined some columns of the database to tell the model that the columns have more significance when combined. This combination acted as a catalyst to boost my models accuracy higher than ever.

Key Decisions while making the Project

1. In the database used, there are so many store names named 'S1,S2,S3 ...' etc. It's the same case with the location of the stores 'L1,L2,....' etc. This uses a lot of computational power which makes the model run slower. This is why i encoded it to numbers(ex: S1=0,S2=1....) so it is easier for the model to compute it. This saves significant time and RAM.
2. I chose LightGBM specifically for its leaf-wise tree growth. LightGBM excels at isolating these highly specific, dense pockets of data, allowing me to capture those nuanced behavioral shifts without overfitting the broader trends.

The Challenges Faced

The biggest challenge was dealing with mathematical redundancy, specifically a phenomenon known as "cannibalization." I was incredibly proud of my custom "Is_Weekend" feature, assuming it would be the ultimate key to predicting sales spikes. However, when I ran the Feature Importance analysis, Is_Weekend was pushed down the list.

Why? Because my model was actually too smart. It realized that my DayOfWeek feature already contained the weekend days (days 5 and 6). DayOfWeek cannibalized the mathematical importance of the weekend flag. I had to learn how to interpret the model's logic and accept that while my custom feature helped fine-tune the margins, the broader 7-day cycle was the true driver.

Key Takeaways from this Project

I learned about the basic to intermediate level of data transformation in this project. This project opened a whole different perspective to look at data for me as I got to know data optimization techniques like categorisation and merging different columns to make more senseful data out of it.

I mean the result speaks of itself, my model achieved a fantastic R2 score of 0.90 with only a measly 6.38 mean absolute error.

```
Final MAE: 6.386628581654325  
Final R2 Score: 0.903529086446564  
CPU times: total: 8min 51s  
Wall time: 34 s
```

This means that my model can explain 90% of the retail demand data with an error of 6.38 orders per 100 orders.

From this project I learned that fitting a model is not only a matter of using the `.fit()` command but also to understand the nature of the data and adapt to it. I learned that during the 'Feature Engineering' part of the project.

I learned about the `astype('category')` feature. This helped to reduce the time taken and computational power required to run the model hence reducing the run time and load on the cpu.